

Class Skill Duration Reduction

Legend of Mushroom — Reverse Engineering Deep Dive #47

Sources: game_script.js engine code + decoded config tables (Skill, Skill_level, Buff, Trap, Relic, Fly_entry, Language_en; XOR key 24455) • Credit: Bird → Discord @birrrd08 • 2026-08-18

The question: Does the avian affix **No Speeding** (“Reduces all enemies' active skill durations by X%”) shorten the **Sacred Hunter's class skill freeze** — Piercing Boneforge's energy-regen lockout — or only “active skill” buffs?

Verdict: YES. No Speeding and Speed Disruption both reduce the Hunter freeze. The engine has no “active skill vs class skill” distinction at the attribute level — every buff or debuff duration a unit's skills produce runs through one shared pipeline. ChatGPT's “it doesn't affect the skill” is wrong; the in-game text listing four counters is right, and the mechanism behind each is decoded below.

1 · TL;DR — The Four Counters

Counter	Source	Mechanism	Hunter freeze?	Range
No Speeding	Avian affix 3102 → skill 23603	Attribute debuff on attr 1061 skillbuff_time_all (permanent aura)	YES	−8% (L1) → −20.8% (L17)
Speed Disruption	Avian affix 3104 → skill 23605	Identical buff chain to No Speeding — hence “cannot stack”	YES	−7% (L1) → −18.2% (L17)
Thorny Spore	Relic 4035 → skill 4035	SKILL_BUFFTIME_ADD (group 360) buff with explicit skill-ID list; permanent aura	YES — 1055 in list	−10% (L1) → −20% (L11 = relic L100)
Dawn Warwing	Back accessory 70405 → skill 18040	Same group-360 mechanism, applied for 60s	YES	L1 −20%, L2 −25%, L3 −30%

Stacking: the two avian affixes are hard-blocked from stacking with each other (literally the same buff, plus conflict_entry on the affix roll). Everything else stacks — attr-1061 sources add *against the base duration*, then each group-360 buff multiplies the result.

Worked example (base 4.0s freeze; No Speeding L15 −19.2%, Thorny Spore max −20%, Dawn Warwing L1 −20%):

$$(4.0 - 4.0 \times 0.192) \times 0.80 \times 0.80 = 4.0 \times 0.808 \times 0.64 \approx 2.07 \text{ s}$$

Not 2.56s — that figure comes from wrongly excluding No Speeding. “Lunar Sprite's −19.2%” is simply No Speeding at level 15 rolled on that avian; it absolutely applies.

2 · The Duration Pipeline (Engine Code)

Every timed effect a skill produces — self-buffs, enemy debuffs, summon lifetimes, trap windows — gets its duration from **one** code path in game_script.js (deobfuscated, simplified):

```
// Called for every buffGroup entry when a skill effect executes:
duration = cast.data.getSkillFactAttrValue(baseTime, useSkill.config.id,
                                           ATTR.active_skillbuff_time);
for (buff of cast.buffCtr.getBuffByType(SKILL_BUFFTIME_ADD)) { // group 360
```

```

    duration = buff.getFixTime(duration, useSkill.config.id);
}
runner.addBuff(target, buffId, duration, coefficient);

// Inside getSkillFactAttrValue(t, skillId, attrGroup):
var n = t;
for (attr of attrGroup) {
    // Step 1: per-skill attrs (MULT)
    f = roundInt(getSkillAttrByAttrId(skillId, attr.id)
        + getSkillAttrByAttrId(0, attr.id));
    n = round(n * round(1 + f / 1e4));
}
if (attrGroup == active_skillbuff_time) // Step 2: attr 1061 (ADD vs BASE)
    n = Math.max(0, n + round(t * this.getAttrib(ATTR.skillbuff_time_all)));
return n;

// BuffSkillBuffTimeAdd.getFixTime (group 360):
getFixTime(t, skillId) {
    if (t == -1) return -1; // permanent buffs untouched
    if (this._skillList.includes(skillId)) // _skillList = buff.param5
        return round(t * (1 - this.skillPar)); // Step 3: MULTIPLICATIVE cut
    return t;
}

```

Three hooks, in order:

- **Per-skill duration attributes** (e.g. Prophet's "prolong active skills by 40%") — multiplicative, keyed to specific skills.
- **Attr 1061** skillbuff_time_all — one number on the **caster**; adjustment is base × attr1061, added to the running result. **No skill filter of any kind.** The `Math.max(0, ...)` floor means duration can never go below zero.
- **Group-360 SKILL_BUFFTIME_ADD buffs** on the caster — each checks its own skill-ID list (param5) and multiplies by (1 – reduction) on match; multiple buffs compound.

Key consequence: the debuffs live on the **caster**. "Reduce all enemies' ... durations" items plant an aura debuff on the enemy unit; when that unit casts *its* skills, its own debuffed attr 1061 / group-360 buffs scale the durations of whatever those skills apply — including debuffs applied **to you**. That is exactly how a duration reducer on your side shortens the freeze the enemy Hunter puts on you.

3 · Anatomy of the Sacred Hunter Freeze

```

Skill 1055 (Piercing Boneforge): skillEffect1 = [10552, 10551]
Effect 10551: buffGroup = [[1, 20042, 1, 4]] <- buff 20042, BASE TIME = 4 s
Buff 20042: action "pause_cd", group 3 (CTR), param1 = 6, mutex 1

```

- **Base freeze = 4.0s at every skill level** — all 220 Skill_level rows for 1055 carry desc_parm [..., 40, 8, 6, 4]; the 4s never scales. Matches the community's "Hunter base interruption: 4.0 sec".
- pause_cd with param1 = 6 pauses energy/cooldown accumulation on 6 active-skill slots — it **stops skill regen entirely** for its duration. The counters shorten the *freeze duration*, not the regen rate.
- The 4s base is an ordinary buffGroup entry of an ordinary skill effect, so it runs the full \$2 pipeline. **Nothing exempts class skills.**
- Related (doc 37): buff 20042 is CTR group 3 → removable by "clear CTR" cleanses; CONTROL_RES does **not** reduce it.

4 · The Four Counters, Mechanism by Mechanism

4.1 No Speeding (avian affix 3102)

```
Fly entry 3102 -> passive skill 23603 (levels 1-17)
Skill 23603: buffGroup = [230040] (passive)
Buff 230040: action "trap", param1 = 1010 (aura carrier)
Trap 1010: duration -1 (permanent), range 10000 (whole arena),
           target = all enemies, buffId [[230041, -1]]
Buff 230041: action "attrib", param1 = 1061 (skillbuff_time_all),
           param3 = -10000, mutex 2
```

A permanent, arena-wide aura planting an **attr-1061 debuff on every enemy**, scaled by skill level (+0.8%/level from 8%; the tooltip's "8%" is level 1). Decoded coefficients:

Level	Reduction	Level	Reduction
1	-8.0%	13	-17.6%
5	-11.2%	14	-18.4%
9	-14.4%	15	-19.2%
11	-16.0%	16	-20.0%
12	-16.8%	17 (max)	-20.8%

The **"Lunar Sprite -19.2%"** seen in the wild is **No Speeding at level 15**. Because it is an attr-1061 debuff it applies to **every** duration the debuffed enemy's skills produce: hero actives, pal-skill buffs, and **class skills** alike.

4.2 Speed Disruption (avian affix 3104)

Skill 23605's buffGroup is [230040] — **the same buff as No Speeding**. Identical chain, only the coefficients differ: -7% (L1) → -18.2% (L17). Two stacking locks: conflict_entry: [3102] on the affix roll, and the shared buff with mutex 2. The tooltip's "cannot stack with No Speeding" is literal — it is the same debuff.

4.3 Thorny Spore (relic 4035)

Correction to docs 24 / 09_RELICS: relic 4035 is Thorny Spore (Language table name id 403285 = "Thorny Spore"; skill desc 403289 = "Reduces the duration of all enemies' Class Skills by ##1%."). The earlier guesses — 4032 = "Thorny Spore?", 4035 = "Nirvana Spore?" — are wrong.

```
Skill 4035: buffGroup = [40351]
Buff 40351: action "trap", param1 = 404
Trap 404: duration -1 (permanent aura), all enemies, buffId [[40352, -1]]
Buff 40352: action "skill_bufftime_add", group 360 (SKILL_BUFFTIME_ADD)
           param5 = [1016,1017,1018, 1053-1058, 1065, 1066, 1067, 1041, 106501]
```

Reduction scales -10% (skill L1) → **-20%** (L11 = relic level 100). Multiplicative, permanent. The target list, with names from Language_en:

ID	Skill	What it is
1053	Blades Reunion	Martial Sage class skill
1054	Shattering Axe	Warbringer class skill
1055	Piercing Boneforge	Sacred Hunter class skill — the freeze
1056	Sun Pursuit	Plume Monarch class skill
1057	Crane's Whisper	Prophet class skill
1058	Galaxy Dive	Darklord class skill
1066	Tamer of Beasts	Beastmaster class skill

ID	Skill	What it is
1067	Wilting Souls	Supreme Spirit class skill
1016	Melon Drop	Hero active (CC)
1017	Cocklebur Dance	Hero active (CC)
1018	Meteor Blitz	Hero active (CC)
1041	Spore Shot	Hero active
1065	Spore Duo	Hero active
106501	Whip of Imprisonment	Hero active

So “Class Skills” in the tooltip is under-descriptive: the reduction also covers six CC/utility **hero actives**. All eight tier-5 class skills are present.

4.4 Dawn Warwing (back accessory 70405)

```
Skill 18040: buffGroup = [180312]
Buff 180312: action "trap", param1 = 757
Trap 757: duration -1, all enemies, buffId [[180313, 60], [180314, 60]] <- 60s!
Buff 180313: action "skill_bufftime_add", group 360,
              param5 = same list as Thorny Spore minus 106501
```

The tooltip's “for 60s” is the debuff duration on trap 757's application — after 60s the enemy's class-skill durations return to normal. Level scaling (decoded, matches desc_parm):

Level	Evasion / Move SPD	Class-skill duration	Debuff window
1	+12%	–20%	60s
2	+16%	–25%	60s
3 (max)	+20%	–30%	60s

The commonly quoted –20% is **level 1**; a maxed Dawn Warwing cuts 30%. *Open item*: trap 757 also applies buff 180314 (attrib on ATK, value 0) for 60s — likely a placeholder; it does not affect the duration math.

5 · Stacking Math

$$\text{freeze} = \max(0, B \times \Pi(\text{per-skill attrs}) + B \times \text{attr1061}) \times \Pi(1 - r360_i) \quad B = 4.0s$$

- **attr-1061 sources** add with each other, computed against the *base*, not the running product. In practice only one applies (No Speeding XOR Speed Disruption).
- **group-360 sources** each multiply the result; Thorny Spore and Dawn Warwing are separate buffs and both apply.
- The **caster's own** attr-1061 *bonuses* (e.g. relic skillbuff_time boosts) add against the same base and can partially offset the debuff.

Setup	Calculation	Freeze
Nothing	4.0	4.00s
Thorny –20% + Dawn Warwing –20% (the “2.56s” claim)	$4.0 \times 0.8 \times 0.8$	2.56s
+ No Speeding L15 (–19.2%)	$(4.0 - 4.0 \times 0.192) \times 0.8 \times 0.8$	≈2.07s
Max everything: NS L17 –20.8%, Thorny –20%, DW L3 –30%	$(4.0 - 4.0 \times 0.208) \times 0.8 \times 0.7$	≈1.77s

The Discord debate, settled

- “ChatGPT says No Speeding doesn't affect the skill.” — **Wrong**. No Speeding is an attr-1061 debuff and attr 1061 has no skill filter.
- *The forwarded in-game text* (“Only the following are capable of countering this effect: Speed Disruption, No speeding, Thorny Spore, Dawn Warwing”) — **Right**, and now explained: two work through attr 1061 (global), two through group-360 skill lists that explicitly include 1055.
- “Lunar Sprite's -19.2% is for active skill duration, so I would not apply it to the Hunter's Class Skill” — **incorrect exclusion**: that -19.2% is No Speeding L15, and it applies. Expected freeze $\approx 2.07s$, not 2.56s.

6 · Corrections & Implications for the Repo

1. **24_RELIC_SYSTEM / 09_RELICS** — relic 4035 is **Thorny Spore** (-10%→-20% enemy class-skill/CC-active durations, permanent aura), not “Nirvana Spore?”; relic 4032 is not Thorny Spore.
2. **34_PVP_META_ANALYSIS** — “no visible counter-mechanic exists in the data” for `pause_cd` is disproven: four counters, two mechanisms, all verified.
3. **V1 simulator** — `affixNoSpeeding` currently scales only active-skill buff durations; it should also scale the class-skill freeze (attr-1061 semantics), with Thorny Spore / Dawn Warwing as separate multiplicative factors — Dawn Warwing expiring at $t = 60s$.
4. **Naming trap** — attribute 1041 is `active_skillbuff_time` while *skill* 1041 is “Spore Shot”: same number, unrelated namespaces. Attr **1061** (`skillbuff_time_all`) is the one the avian affixes debuff.

7 · Confidence & Open Questions

High confidence (read directly from engine code + config data): pipeline order, attr-1061 additivity vs base, group-360 multiplicativity and skill lists, base 4s freeze, all level scalings, the shared No Speeding / Speed Disruption buff.

Not yet verified in-game: frame timing of aura application at battle start (whether a class skill cast in the first instants could precede the aura landing), and the purpose of Dawn Warwing's companion buff 180314.